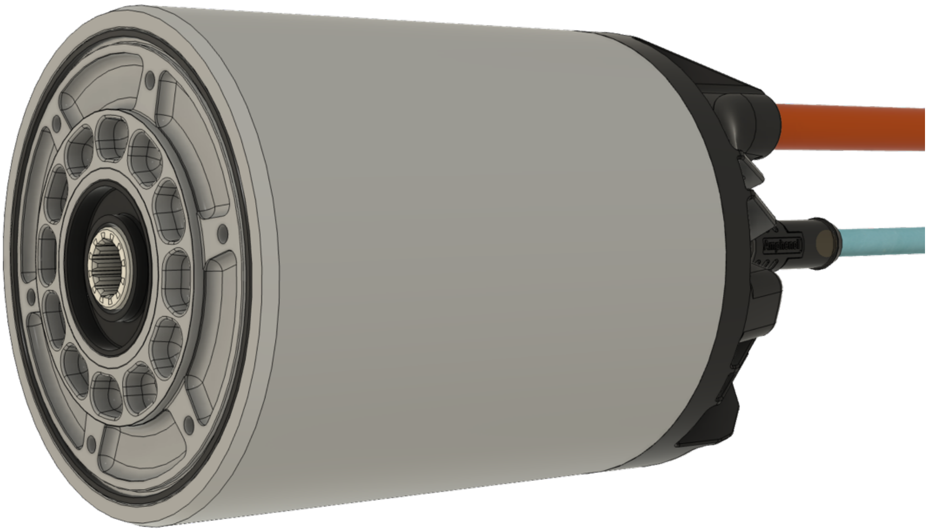
DTI FORMULA STUDENT MOTOR

Application manual for F-MOT motor

Version 1.3

Valid for F-MOT-A motor



Please read the user manual carefully before using the motor. Electrical and mechanical systems can cause danger to humans, property and nature; therefore, precautions shall be taken to avoid any risk.

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1. HISTORY AND RELATED DOCUMENTS

1.1 Document history

10/2024	**V1.0** Document created; basic specifications added
02/2025	**V1.1** Document release • Liquid cooling technology information added, mechanical parameters added • AC shoe terminal type changed
05/2025	**V1.2** Document edit • Position sensor connector pinout modification
06/2025	**V1.3** Document Edit • Pinout description change

2. LIABILITY AND SAFE USE OF THIS UNIT

DTI hardware, DTI CAN Tool and the DTI firmware are experimental products designed to develop, evaluate and test electrical systems incorporating electric motors or actuators. Electrical systems can cause danger to humans, property and nature; therefore, precautions shall be taken to avoid any risk. Under no circumstances shall the device be used where humans or property are put to risk without thoroughly validating and testing the whole system. Software and hardware interact in various ways, and developers cannot foresee all possible combinations of hardware used together with software, nor problems that can occur in these different combinations. Can Tool and the DTI firmware are experimental software designed to develop and test. Electrical systems can cause danger to humans, property and nature; therefore, precautions shall be taken to avoid any risk.

Only trained and educated personnel are authorized to configure, integrate and operate the device who know the potential risks, and can act in case of any misbehavior, or fault that occours.

The integrator has the responsibility to select the proper hardware components for the specific use-case. DTI takes no responsibility for any issues, problems, failed examinations, failed registration procedures or evaluations due to inappropriate configuration, system design or general system structure or lacking documents, certifications for the specific area-of use.

Things that can happen, even when using the correct settings, are

- electrical failure
- fire
- electric shock
- hazardous smoke
- overheating motors and actuators
- overloaded power sources, causing fire or explosions (e.g., Lithium-Ion Batteries)
- motors or actuators stopping from spinning/moving
- motors or actuators locking in, acting like a brake (full stop)
- motors or actuators losing control over torque production (uncontrolled acceleration or braking)
- interferences with other systems
- other non-intended or unforeseeable behaviour of the system

DTI CAN Tool and the DTI firmware are developer software that for safety reasons may only be used

- by experts and experienced users, knowing exactly what they do.
- following safety standards applicable in the area of usage.
- under safe conditions where software or hardware malfunction will not lead to death, injuries or severe property damage.
- keeping in mind that software and hardware failures can happen. We can't provide any kinds of warranty because every system is unique and we cannot make sure its safety and integrity. Although we design our products to minimize such issues, you should always operate with the understanding that a failure can occur any time without warning. As such, you shall take the appropriate precautions to minimize danger in case of failure.

Each and every motor is being tested as a part of the manufacturing progress. DTI assumes no liability in case a customer uses system components for the purposes that they have not been developed for or tested.

DTI reserves the right to change any information within the scope of this whole manual. All connection circuitry described is meant for reference only, are theoretical circuits and not mandatory. DTI does not assume any liability, expressively or inherently, for the information contained in this document, for the functioning of the device or its suitability for any specific application.

3. WARNING SIGN EXPLAINED

The Warning sign contains the most important safety notices that has to be taken into consideration by everyone who approaches the inverter. The sign cannot be removed or covered off he controller.

In case of the sign is being damaged, or detached from the inverter, please get in touch with DTI as soon as possible at the availability mentioned on the cover page.

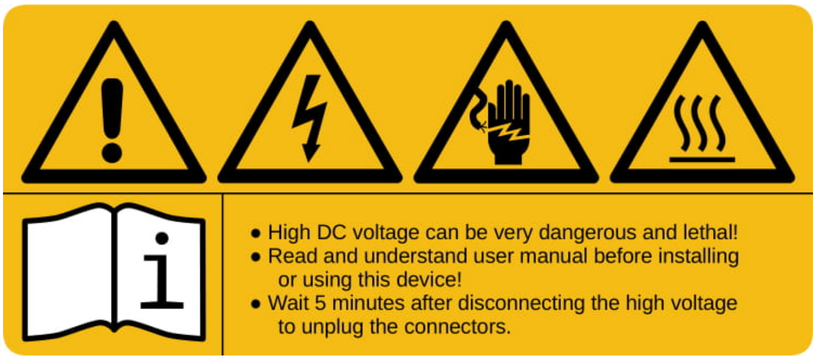
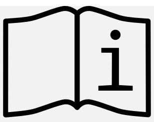


Figure 1. Safety note

Table 1. Safety notes explained

Description	
	Please read this manual carefully, with particular regard to safety notes.
	**Danger of electric shock** Touching HV wires and HV connectors are prohibited without ensuring that the HV is disconnected.
	**Danger of high voltage** The device is attached to high voltage.
	**Danger of hot surfaces** The inverter case, and /or the coolant fluid can get hot during and after operation, touching any of these can cause burn damage.
	**General Warning** Only trained and educated personnel can do integration and maintenance.

4. OVERVIEW

Key Features[1]

- Peak / Nominal Torque: 31,6 Nm / 24,6 Nm
- Peak / Nominal Current: 66 Arms / 22,6 Arms
- Maximum Speed: 20,000 rpm
- Peak Power: 35 kW
- Peak Efficiency: >96%
- Motor Weight: 3,7 kg
- 94.5 mm diameter, 113 mm length (without plastic end housing)
- Position Sensor: TE connectivity 2-1393047-5
- Temperature Sensor: PT1000

Application

- Formula Student 2/4WD drives
- Prototype small cars, motorcycles, karts
- Battery powered machines

Description

The F-MOT-A is a high-performance motor specifically designed for Formula Student teams, developed with the latest competition rules in mind. It combines optimal power delivery with a lightweight design, making it ideal for high-speed racing applications.

The motor is equipped with a detachable high-voltage cable, which simplifies vehicle assembly and makes the motor installation and removal process easier.

[1] Data provided by Fischer Elektromotoren.

5. ELECTRICAL CONNECTIONS

5.1 Connector identification

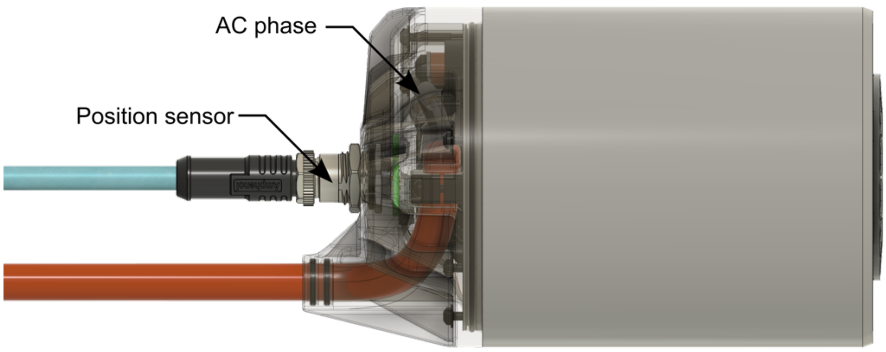


Figure 2. Electric connector identification

Table 2. Electric connection

Pin name	Pin description
AC phase	three cable lugs for AC connection
Position sensor	M12 circular connector for position sensor and thermometer

5.2 Position sensor connector pinout

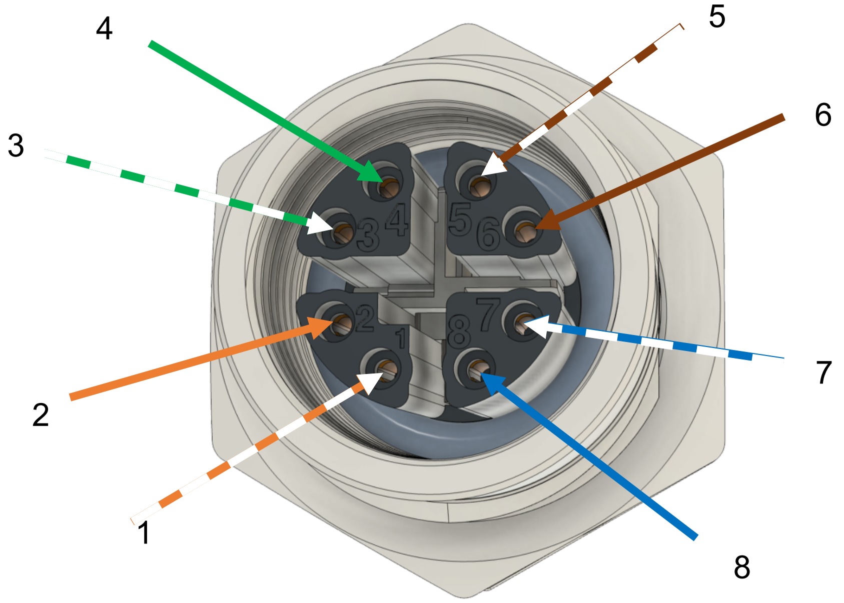


Figure 3. Position sensor connector (Id and wire color)

Connector type: Amphenol **MSXS-08PFFP-SF7002**

Recommended cable: Amphenol **MSXS08ML-SXSML-SXXXX**

Table 3. Position sensor connector pinout description

Id	Wire Color	Pin description
1	WHT/ORANGE	Res. Cos Signal +
2	ORANGE	Res Cos Signal -
3	WHT/GREEN	Res Sin signal +
4	GREEN	Res Sin signal -
5	WHT/BROWN	PT1000 signal Pin 1
6	BROWN	PT1000 signal Pin 2
7	WHT/BLUE	Res Exc Signal +
8	BLUE	Res Exc. signal -

It is important to use shielded cables, and the internal wires of the cable should be twisted in pairs. The twisted wire pairs should be routed together with their corresponding complementary pairs to minimize electromagnetic interference.

5.3 AC phase connector pinout

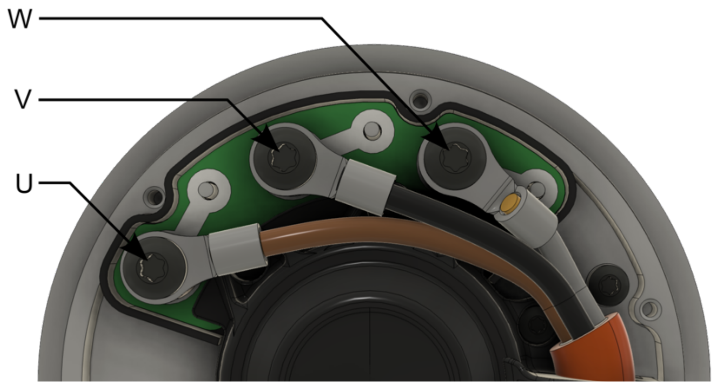


Figure 4. AC phase sensor connector

Connector type:	TE Connectivity **321827**
Heat shrink	TE Connectivity **ATUM-8/2-0-SP**
Recommended cable:	Coroflex **9-2641 (3 x 2,5 mm2)**

Table 4. AC phase connector pinout description

Id	Pin name	Pin description
1	U	Motor phase wire U
2	V	Motor phase wire V
3	W	Motor phase wire W

5.4 AC High voltage connection

The thread size on the connection surface is M5. A cable lug must be used that can handle the AC current of the motor.

> The maximum tightening torque of the screw is 2.2Nm.
> Use a shrink tube to ensure that the connection is properly isolated.
> Thread locker fluid is recommended.

The phases connector is secured with T20 Torx screws. The recommended tightening torque of the screw is 2Nm.

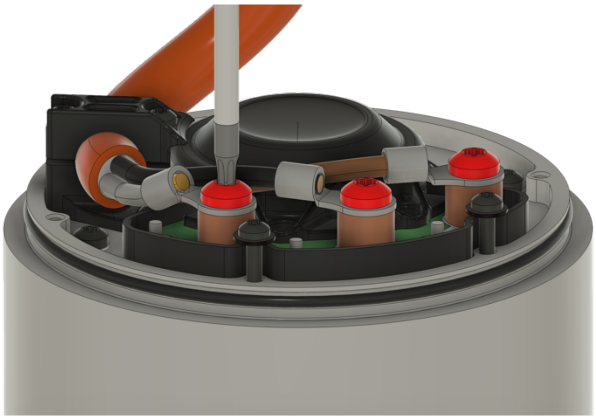


Figure 5. Motor phase fixing

After tightening these screws to the specified torque, the cable can be fixed using the cable fixing component, which is held in place by two screws.

> Secure the cable in such a way that it cannot rotate with moderate force, as this could be hazardous due to high levels of vibration.
> Thread locker fluid is recommended.

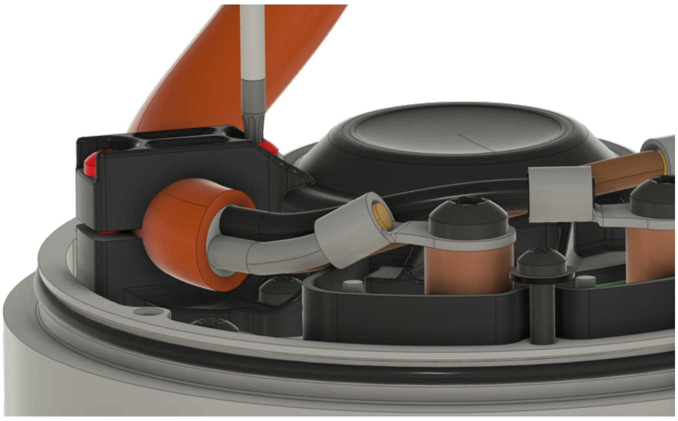


Figure 6. AC cable fixing

Connect the Resolver position sensor and the motor temperature sensor wires to the position sensor PCB

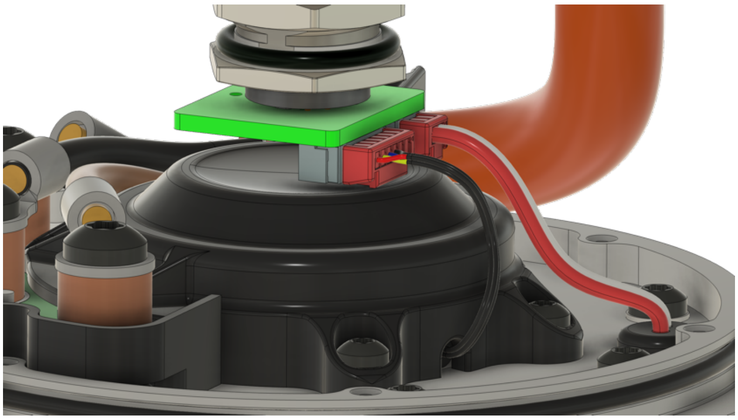


Figure 7. Sensor PCB connection

Place the two AC wire sealings to the cap and the cap sealing to the motor. Use two pieces 10mm x 2mm O-rings to seal the cable and one 82mm x 2mm for the cap sealing.

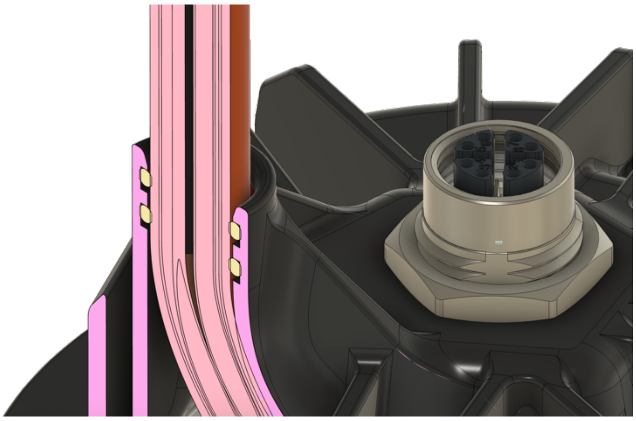


Figure 8. AC cable sealing

Position the plastic cap in place. Please use the 6 pieces M2.5 screws and secure the retaining nut of the motor’s circular connector.

> Thread locker fluid is recommended.
> Ensure the tightening torque does not exceed 0.7 Nm.

Figure 9. Plastic cap fixing

5.5 Grounding

> The motor casing must be electrically connected to the chassis **which is the potential of the low voltage ground**.

All wires connecting to the motors have their shielding grounded on the inverter side. To avoid ground loops, the shielding for the AC phase wires and the motor position sensor wires is not connected on the motor side. The motor can establish an electrical connection to the upright, and through the suspension arms, to the vehicle chassis. This connection should be replaced with a dedicated wire between the common grounding point and the motor / upright.

6. LIQUID COOLING

As this design is intended for integration into a highly specialized wheel hub, each team is responsible for developing its own cooling solution. Detailed material specifications are included in the motor specification. The wall thickness between the stator iron and the motor exterior is 4.6 mm. If additional cooling capacity is required, a sleeve can be fitted onto this surface to enhance the heat dissipation area.

Due to manufacturing technologies, there are zones where the mounting or sealing shall not be placed. The laser welded surfaces may not be even enough to apply sealing, therefore it should be kept in or out of the liquid cooling zone. The below chart is for reference only. For detailed information please refer to the cooling surface drawings.

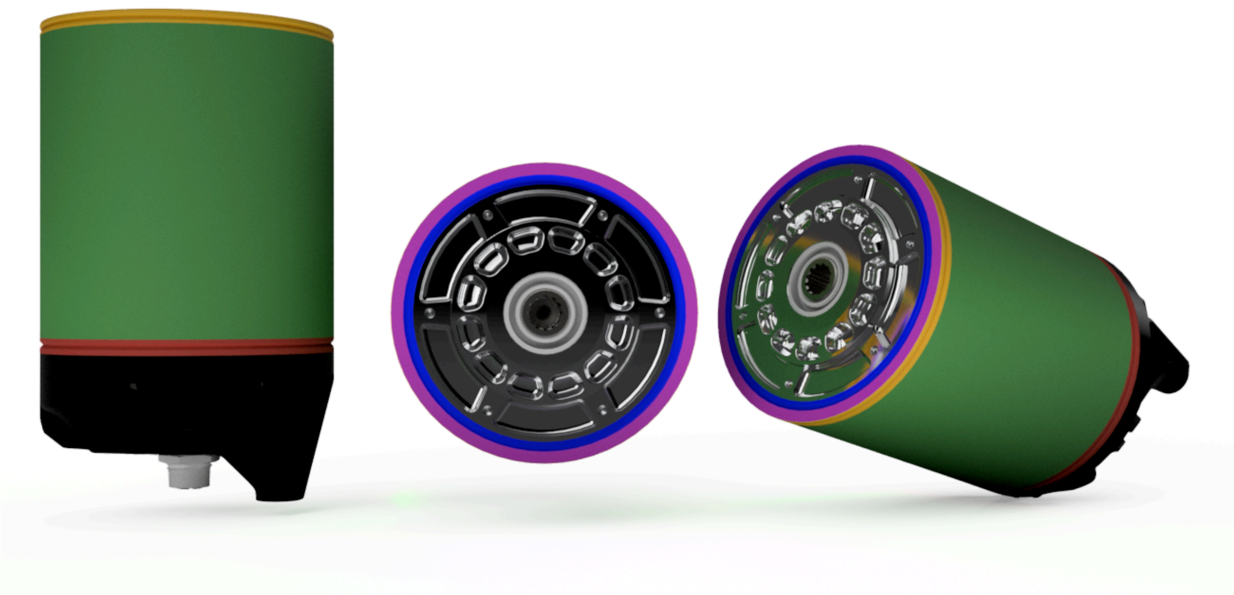


Figure 10. Cooling zones

Table 5. Cooling zone color coding

Colour coding	Description
Green	Recommened cooled zone *(can seal here)*
Red	Outside cooled zone *(can not seal here)*
Yellow	Optionally part of cooled zone *(can not seal here)*
Magenta	Optionally part of cooled zone *(can seal here)*
Blue	Sealing nest for O-ring
Black	Plastic cap, outside cooled zone *(can not seal here)*
Grey/Metal	Metal parts, outside cooled zone *(can not seal here)*

Dti Team developed a simulation software that assists teams in evaluating the thermal dissipation characteristics of the inverter and motor. The primary heat-generating factor for the motor is its rotational speed, which increases iron losses, while the AC current contributes to conductive losses. The DTI CAN Tool feature further supports Formula Student teams by providing insights into the thermal behavior of both the inverter and motor through transient simulations. It also allows users to determine the required coolant temperature for varying load cycles.

To use this feature, follow these steps:

- Download the latest DTI CAN Tool graphical interface from our website.
- After installation, you do not need to connect the DTI diagnostic tool to the COM port; simply close the pop-up window and select the "Simulation" function from the left menu bar.
- Further instructions will then be visible

It is important that the uploaded CSV file matches the format, number of columns, and units as per the provided template. The CSV file can contain negative current values, but they must not exceed the inverter's maximum rating. If any value exceeds the maximum rating, it will be capped.

If the inverter temperature exceeds the limit value, the inverter will begin to limit, and any AC current values above that threshold will no longer be achievable. If the AC current decreases again, operations will resume without limitations.

7. MECHANICAL DRAWINGS

Dimensions are shown in mm.

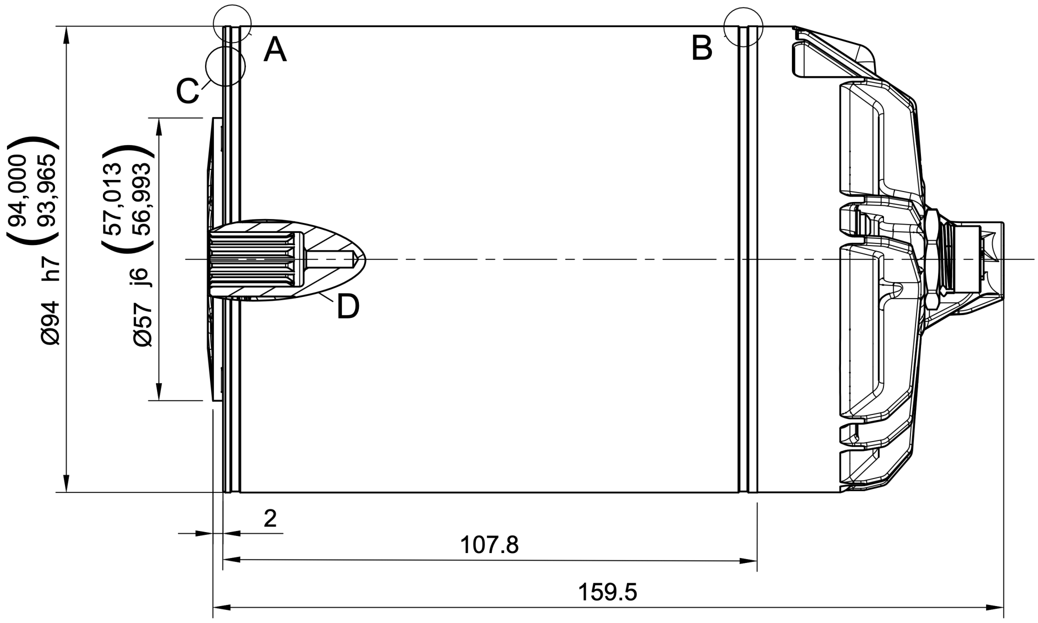


Figure 11. Top View

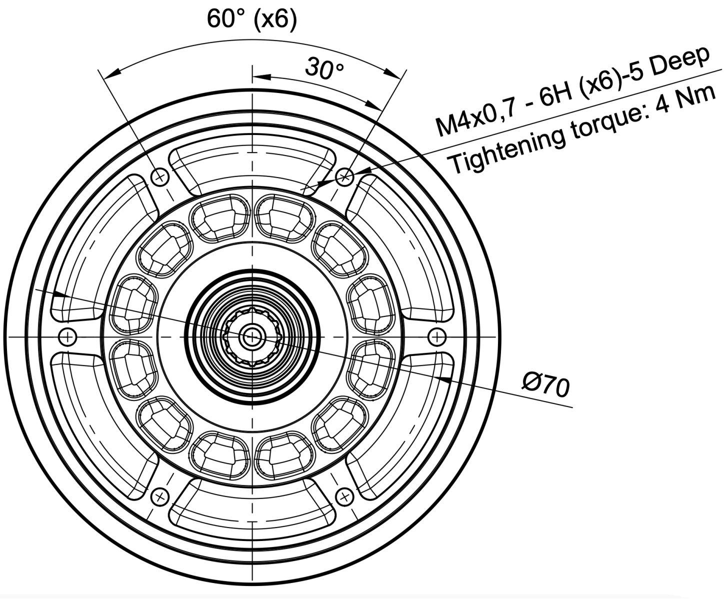


Figure 12. Front View

For a more deatailed view incuding extended parameter description please refer to the drawing or CAD model appendix.

8. LIST OF TABLES

9. LIST OF FIGURES